Aplysia R15 Reading Map

MATH3888: experimental baseline -> model -> bifurcation analysis -> extensions

Suggested order **1 Adams & Benson (skim) -> 2 Junge & Stephens 1973 -> 3 Plant & Kim 1975 -> 4 Plant & Kim 1976 -> 5 Plant 1981**

Then: Adams 1985 + Adams-Levitan 1985 for later physiology; Faber-Klee/Eaton for supporting history; Levitan 1988 for a serotonin extension.

FOUNDATION + MAIN MODEL

CORE

Junge & Stephens (1973) - Cyclic variation of potassium conductance

Contains: Primary R15 experiments: post-burst / TTX slow-wave hyperpolarization, conductance and K+ dependence.

Use: Best accessible experimental foundation for the cyclic K+ mechanism used by Plant & Kim.

CORE

Plant & Kim (1975) - Mechanism underlying bursting in R15

Contains: TTX slow-wave model; fast-slow reduction to 2D; nullclines and stable relaxation oscillation.

Use: Understand the original mathematics and reproduce the TTX slow-wave / current-response baseline.

CORE

Plant & Kim (1976) - Modified Hodgkin-Huxley bursting model

Contains: Fuller conductance-based R15 model with explicit functions/parameters; normal bursting and TTX behavior.

Use: Most useful computational starting point if the project models full bursting rather than only the slow wave.

CORE

Plant (1981) - Bifurcation and resonance in a bursting model

Contains: Explicit analysis of a parameter-driven transition between periodic waveforms; resonance bifurcation.

Use: Direct template for the MATH3888 continuation/bifurcation component and for choosing meaningful parameters.

SUPPORT

Faber & Klee (1972) + Eaton (1972)

Contains: Early R15 electrophysiology: burst waveform/current-voltage features; Eaton tests delayed outward-current decrease and K+ accumulation.

Use: Historical experimental support and useful competing/extra mechanisms for the discussion/limitations section.

LATER PHYSIOLOGY + EXTENSIONS

START

Adams & Benson (1985) - R15 review + comprehensive model

Contains: Large synthesis of R15 morphology, currents, bursting, modulation, prior models, and a data-grounded computer simulation.

Use: Skim first as the literature map; return to its ionic-model sections when interpreting later experiments.

CORE

Adams (1985) - Slow depolarizing and hyperpolarizing currents

Contains: Primary voltage-clamp experiments identify ID and IH with very different time scales; includes a bursting simulation.

Use: Strong later experimental evidence for the currents/timescales that generate and terminate bursts.

TARGET

Adams & Levitan (1985) - Voltage and ion dependence of ID and IH

Contains: Measures activation, voltage range and ionic dependence of the slow currents identified by Adams.

Use: Use to refine/critique model terms and justify which ions/conductances are physiologically plausible.

EXTEND

Levitan & Levitan (1988) - Serotonin/cAMP modulation

Contains: Primary experiments: 5-HT changes R15 bursting via cAMP, modulating K+ and subthreshold Ca2+ currents.

Use: Natural drug/neurotransmitter extension: ask how parameter changes alter bursting or create tonic spiking.

NOTE

Mathieu & Roberge (1971) - inaccessible full paper

Contains: Key TTX current-injection traces are reproduced in Plant & Kim (1975); the original is useful but not essential.

Use: Not a blocker: cite the original experiment, but base figure-specific analysis on the accessible Plant & Kim reproduction.

PROJECT PATH **Experiment (Junge / Adams) -> reproduce Plant-Kim -> equilibria + stability + nullclines -> PyDSTool continuation / periodic orbits -> compare with traces -> extension (serotonin, later currents, or alternative mechanism)**